

MODELLING HOST-PATHOGEN INTERACTIONS USING MULTISCALE MD SIMULATIONS: THE MYCOBACTERIUM TUBERCULOSIS CASE

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ABSTRACT

Mycobacterium tuberculosis, the causative agent of tuberculosis, possesses an unusually thick, waxy cell envelope composed of a diverse array of complex lipids (1). These lipids not only provide the structural framework of the mycobacterial envelope but also contribute to virulence by perturbing host membranes and interacting with host-cell receptors. Understanding their organization and dynamics is therefore essential for deciphering the molecular basis of mycobacterial pathogenicity.

In this presentation, I will highlight our recent works using multiscale molecular dynamics (MD) simulations to study mycobacterial lipids. Atomistic simulations have provided molecular-level insights into the organization of the outer membrane, the mycomembrane (2). Coarse-grained simulations have then been employed to investigate the redistribution of the major virulence lipid phthiocerol dimycocerosate (PDIM) within the mycomembrane (3). In parallel, all-atom MD simulations have been used to characterize the structure and dynamics of the mycobacterial inner membrane (MIM) (4).

Beyond their role in the bacterial envelope, mycobacterial lipids display distinct behavior in host-cell membranes. When incorporated into host-like membrane models, they disrupt membrane organization, providing a molecular basis for their activity as virulence factors (5,6).

Together, these studies demonstrate how multiscale MD simulations provide a powerful framework for uncovering the structure, dynamics, and biological functions of mycobacterial lipids.

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